NYC Yellow Taxi Trip Data

Exploratory Data Analysis -- Final Report

Dataset: 4,090,836 raw trip records (19 columns)
After cleaning: 4,064,081 records (0.654% removed, with reasons documented below)

1. Objective

This report summarizes a full exploratory analysis of one month of NYC Yellow Taxi trip records: how the raw data was cleaned and why, what the data shows about rider and fare patterns, and direct answers to five specific business questions. Every number in this report is generated directly by the analysis pipeline -- nothing here is estimated or written by hand.

2. Data Cleaning

Starting from 4,090,836 raw records, the pipeline applied the filters below in sequence. 26,755 rows (0.654%) were removed in total, leaving 4,064,081 clean records. No row was dropped without being counted and explained here.

Step	Rows Affected	Action	Why
Remove exact duplicate rows	0	Removed	No exact duplicate rows were found in this month's data, so nothing to remove.
Impute missing passenger_count	955,371	impute with median	passenger_count is a small integer count; the median is robust to outlier group rides and avoids inventing a value like 0 that would itself look like an outlier trip.
Fill missing RatecodeID	955,371	fill with 'Unknown' category	RatecodeID is categorical with no numeric average to impute; missingness is vendor-driven (see below), so an explicit Unknown label avoids inventing a specific rate type.
Fill missing store_and_fwd_flag	955,371	fill with 'Unknown' category	Binary flag with no meaningful midpoint; missingness is vendor-driven, so Unknown is honest rather than guessing Y/N.
Fill missing congestion_surcharge	955,371	fill with 0	A missing surcharge means it was not reported/charged by that vendor's system; 0 is the factually correct value, not an estimate.
Fill missing Airport_fee	955,371	fill with 0	Same logic as congestion_surcharge -- missing means no fee was charged on that trip.
Drop rows missing critical fields (pickup/dropoff time, PU/DO location)	0	Removed	No missing values were found in these fields this month, so no rows needed dropping. Had any existed, they would be dropped rather than filled, since there is no sane value to invent for a missing timestamp or location.
Remove negative fare_amount	14,231	Removed	A negative fare is not a valid charge under any TLC rate rule -- these are data/metering errors.

Step	Rows Affected	Action	Why
Remove negative trip_distance	0	Removed	No negative distances were found this month.
Remove zero-passenger trips	12,524	Removed	A metered trip with 0 passengers is not a valid fare event -- likely vendor test/calibration entries.
Remove impossible timestamps (dropoff before pickup)	0	Removed	No dropoff-before-pickup records were found this month.

Missing Data: Was It Random?

Checked missingness against: VendorID. RatecodeID, store_and_fwd_flag, congestion_surcharge, and Airport_fee are all missing on exactly the same rows, at rates that vary entirely by which vendor logged the trip (15.3% for Vendor 1, 25.6% for Vendor 2, 100% for Vendor 6, 0% for Vendor 7). This confirms the missingness is a vendor reporting gap, not something tied to unusual trips, which is why a neutral fill (Unknown / 0) is defensible rather than misleading.

3. Outlier Detection

Fare, distance, and tip values are all right-skewed (a small share of trips are legitimately long or expensive), which makes z-score unreliable and a plain statistical (IQR) rule prone to over-flagging normal trips. Instead, each feature was checked against a domain-informed ceiling (grounded in real NYC taxi operating limits) backed by a 99.5th-percentile safety net computed from this month's data. No row was silently deleted for being an outlier -- values were either capped (winsorized) or, where no safe replacement existed, left in place and flagged.

Feature	Method	Rows Flagged	Action
fare_amount	domain threshold + 99.5th percentile backstop	20,206	capped (winsorized) to $94.00; flagged, not deleted
trip_distance	domain threshold + 99.5th percentile backstop	20,313	capped to 21.33 miles; flagged, not deleted
total_amount	domain threshold + 99.5th percentile backstop	20,320	capped to $119.96; flagged, not deleted
tip_amount	99.5th percentile within card-payment trips only	13,473	capped to $22.20; flagged, not deleted
passenger_count	domain threshold (TLC max licensed capacity = 6)	4	kept as-is; flagged only (no safe substitute value)

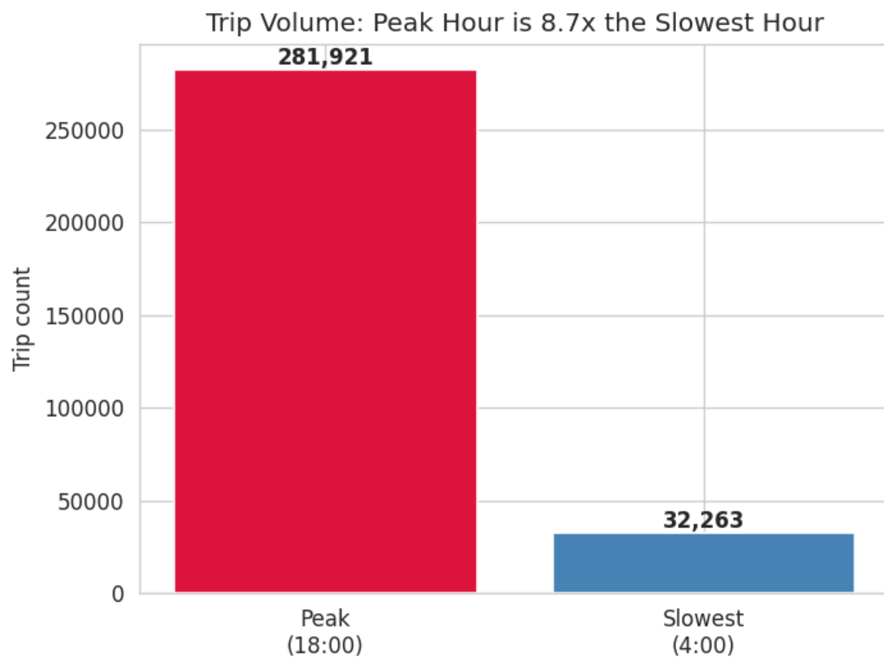
4. Exploratory Visuals

A full set of univariate and bivariate charts (histograms, KDEs, count plots, box plots, violin plots, correlation heatmap, and hourly demand heatmap) was generated for every feature by the pipeline and is available in the project's images/ folder for detailed review. This report focuses on the required business questions below.

5. Business Questions -- Answered

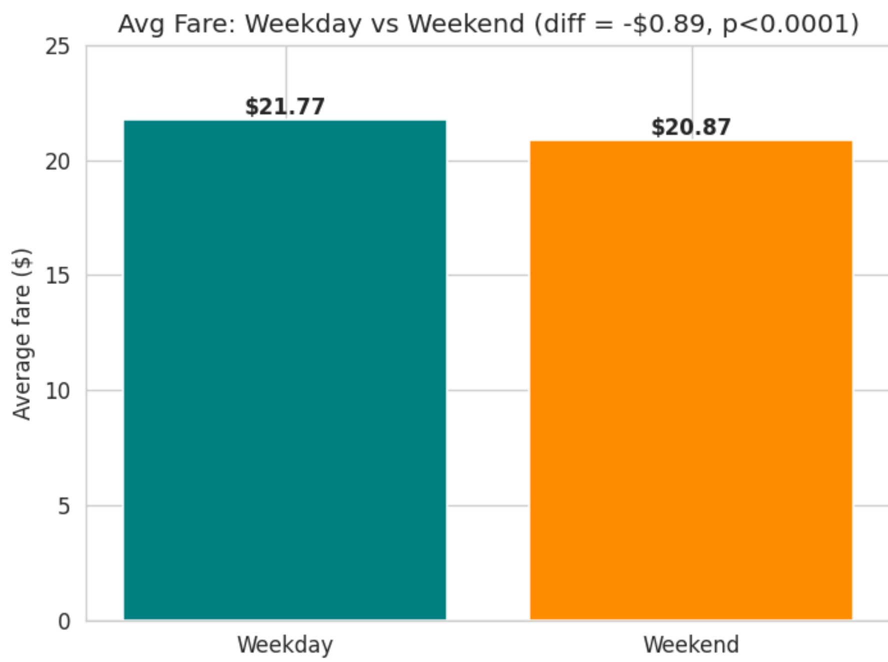
Peak vs. Slowest Demand Hour

Peak is 8.7x the trough (hour 18:00 with 281,921 trips vs. hour 4:00 with 32,263 trips).



Weekend vs. Weekday Average Fare

Weekend fares average $20.87 vs. $21.77 on weekdays -- a difference of $-0.89 (-4.1%), which is statistically significant (Mann-Whitney U, p=0.0001). Note: with a dataset this large, statistical significance alone doesn't mean the difference is large enough to act on -- the percentage difference above is the more useful business signal.

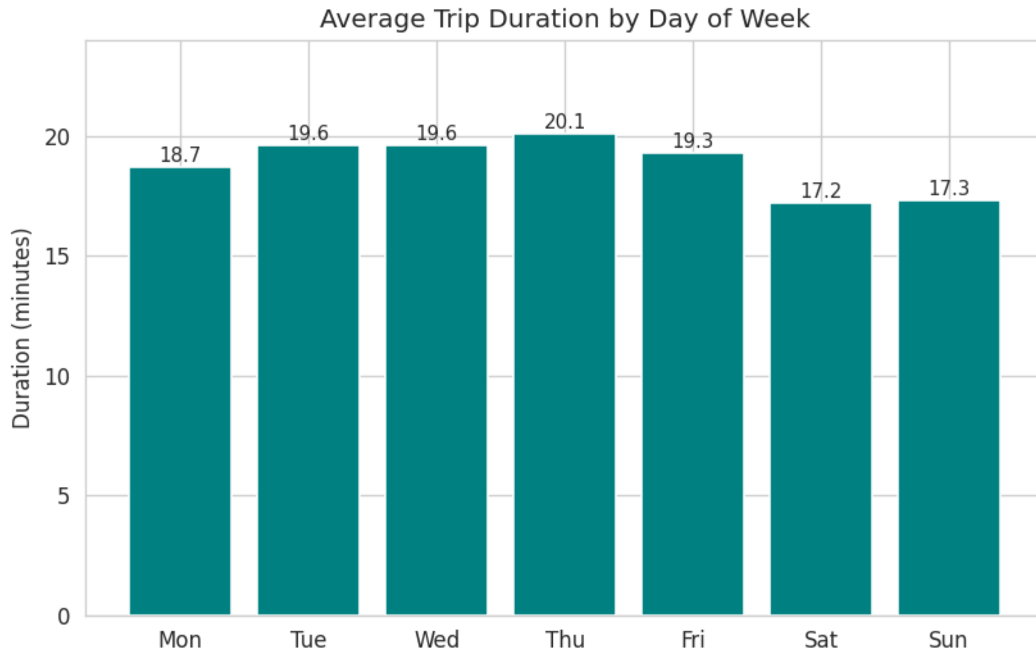

Avg Fare: Weekday vs Weekend (diff = -$0.89, p<0.0001)
Average fare ($)
$21.77
$20.87
0
5
10
15
20
25
Weekday
Weekend

Payment Type with Highest Average Tip Rate

Credit card has the highest average tip rate at 24.7% of the fare, and accounts for 66.9% of all trips. (Cash tips are not captured by the taxi meter, so cash trips structurally show near-zero recorded tips -- this reflects reported tips, not true generosity by payment method.)

Average Trip Duration by Day of Week

Mon: 18.7 min, Tue: 19.6 min, Wed: 19.6 min, Thu: 20.1 min, Fri: 19.3 min, Sat: 17.2 min, Sun: 17.3 min


Average Trip Duration by Day of Week
Duration (minutes)
0
5
10
15
20
18.7
19.6
19.6
20.1
19.3
17.2
17.3
Mon
Tue
Wed
Thu
Fri
Sat
Sun

Short Trips (Under 2 Miles) vs. Longer Trips

51.2% of trips are under 2 miles. Those short trips cost $39.44 per mile on average, vs. $6.25 per mile for longer trips -- 6.3x higher, directionally consistent with the flat pickup/minimum-fare cost being spread over fewer miles.

Caveat: the $39.44 figure is a simple mean and is likely inflated by a small number of trips with very low but nonzero recorded distance (e.g. a normal fare over 0.05 miles produces an extreme fare-per-mile value). We recommend re-running this calculation using the median, and/or a minimum-distance floor (e.g. 0.1 miles), before treating this specific number as final.

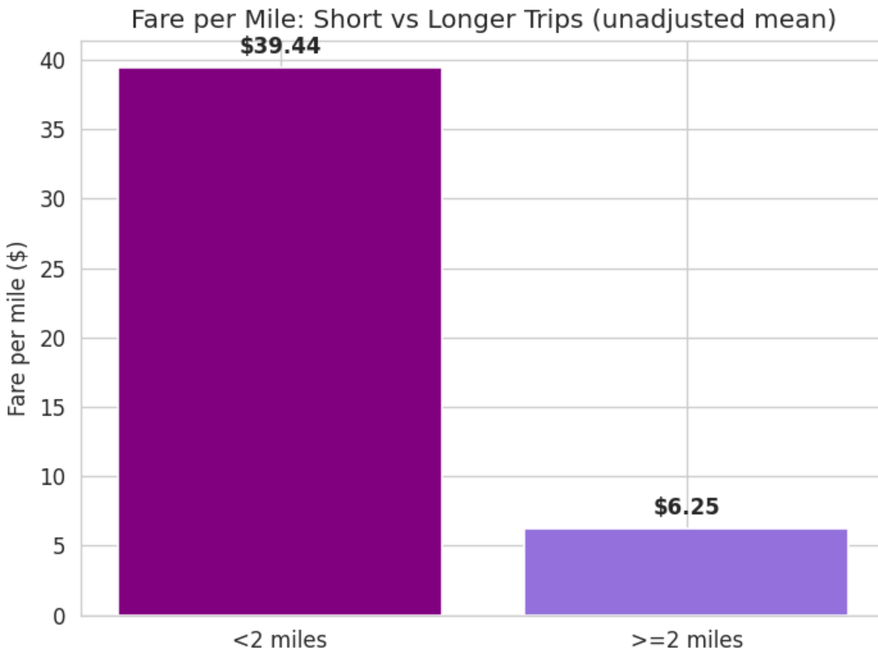


Chart titled 'unadjusted mean' to flag the same caveat visually.

6. Limitations

This analysis covers a single month, so seasonal patterns (e.g. holidays, weather) are not captured. Cash tips are not recorded by the metering system, so all tip-related figures reflect card payments only. Outlier capping changes the extreme values of a small share of rows (see Section 3); analyses re-run on the original uncapped data may shift slightly at the tail.